SCX-Audited Medical Diagnosis:

Multi-Physician Consensus for Certified Clinical Decision-Making

SCX

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Abstract

Abstract. Medical diagnosis is one of the highest-risk domains of human decision-making: the misdiagnosis rate in clinical practice is reported as high as 5–15%, and missed diagnosis of rare diseases is even more common due to low prior probability. This paper formalizes medical diagnosis as a **multi-expert consensus audit problem**, placed under the SCX framework. In this formalization, each physician is an expert f_m, producing a diagnostic opinion $\hat{d}_m$; imaging, laboratory testing, and pathology constitute **multi-modal experts**, each observing the patient's status from different information channels; the second opinion corresponds to $M = 2$; the multidisciplinary team / tumor board is the institutional embodiment of the Yajie consensus mechanism.

We prove three core theorems, each with complete proof (■ **[Full Proof]**).

Theorem 1 (Multi-Physician Misdiagnosis Detection Theorem): For M independent physicians with effective multiplicity $M = M/(1 + (M - 1)\bar{\rho})$, where $\bar{\rho}$ is the average inter-physician error correlation, the probability that all M physicians collectively miss a diagnosis is bounded by $\exp(-2M\,\Delta^2)$, where Δ is the diagnostic detection margin.

Theorem 2 (Multi-Modal Expert Consensus Theorem): When K diagnostic modalities (imaging, laboratory, pathology, genomics) provide conditionally independent diagnostic signals, the Yajie weighted consensus reduces the misdiagnosis probability exponentially in K. We derive the optimal modality weighting scheme: when modality k has diagnostic odds ratio DOR_k, the weight $\omega_k \propto \ln(\mathrm{DOR}_k)$ is optimal in the sense of minimizing consensus error risk.

Theorem 3 (False Negative and Rare Disease Unidentifiability Theorem): When the disease prior probability $\pi_0 \to 0$ (rare disease regime), a negative result from a single diagnostic modality cannot distinguish "the test is correct—the disease is truly absent" from "the test fails—the rare disease is missed." We prove the identifiability gap: $\left|\mathbb{P}(d = 1 \mid \hat{d} = 0) - \pi_0\right| \leq \frac{1-\mathrm{Sens}}{\mathrm{Sens} + \mathrm{Spec} - 1}$, and $M > M_{\min} = \frac{\ln(1/\varepsilon)}{2\Delta^2}$ independent experts are required to shrink this gap below ε.

This paper is not a survey and does not claim to replace physicians. It provides theorems and proofs.

Keywords: medical diagnosis, multi-physician consensus, SCX audit, misdiagnosis rate, Hoeffding inequality, rare disease unidentifiability, multi-modal diagnosis, tumor board, Yajie consensus, Cercis score, Spring gating, diagnostic odds ratio.

1 Introduction

Medical diagnosis is a decision under uncertainty with life-or-death consequences. Despite centuries of clinical experience and decades of evidence-based medicine, the fundamental structure of diagnostic error remains poorly quantified. Published estimates of misdiagnosis rates range from 5% in radiology [1] to 10–15% in general internal medicine [2], and up to 20–30% in emergency department settings [3]. Autopsy studies reveal that 8–24% of causes of death were missed during life [4]. These are not failures of individual physicians — they are structural properties of single-expert decision-making.

The clinical community has long intuited the remedy: get a second opinion. The second-opinion practice ($M = 2$) reduces diagnostic error rates by 15–30% in radiology [5] and changes management in 10–62% of cases across specialties [6]. Tumor boards (multidisciplinary team meetings) — formal meetings where surgeons, medical oncologists, radiation oncologists, radiologists, and pathologists jointly review cases — have been shown to alter diagnosis or management in 25–52% of cases [7]. These practices are instantiations of a mathematical principle that the clinical literature has never explicitly recognized: **multi-expert consensus with independent information channels produces exponentially decreasing error probability**.

This paper provides the missing mathematical foundation. We formalize medical diagnosis as an SCX audit problem where each physician is an expert, each diagnostic modality (imaging, laboratory, pathology, genomics) is an independent information channel, and the clinical decision is a YAJIE consensus over multiple expert opinions. We prove three theorems that establish:

(i) How many physicians are needed to guarantee a misdiagnosis probability below a target ε, given the inter-physician error correlation $\bar{\rho}$ and detection margin Δ (Theorem 1).

(ii) How multiple diagnostic modalities combine to produce a consensus diagnosis whose error probability decays exponentially in the effective number of independent modalities K (Theorem 2).

(iii) Why rare diseases create a fundamental unidentifiability between false negatives and true absence, and how many independent diagnostic modalities are required to resolve it (Theorem 3).

The core mathematical analogy between clinical medicine and the SCX audit framework is summarized in Table 1.

Table 1: Core mathematical analogy: clinical medicine $\leftrightarrow$ SCX audit framework

Clinical Concept	**SCX Analog**
Physician	Expert f_m
Second opinion	$M = 2$ experts
Tumor board / MDT	YAJIE consensus
Misdiagnosis	Noise (Theorem 1 detection)
Imaging / Lab / Pathology	Multi-modal experts
False negative in rare disease	Theorem 3 unidentifiability
Clinical guidelines	Assumption declarations
Diagnostic odds ratio	Expert weight
Patient history	SPRING permanent memory

What this paper is not. This is not a clinical guideline. It is not a systematic review of diagnostic accuracy studies. It does not claim that physicians can be replaced by algorithms, nor that mathematical formalization captures all nuances of clinical reasoning. It is a mathematical paper about multi-expert consensus, error detection probability bounds, and unidentifiability in diagnostic decision-making. The theorems and proofs are the contribution.

2 Formalization: Medical Diagnosis as Multi-Expert Audit

2.1 Patient, Disease, and Diagnosis

Definition 2.1 (Patient State). *A **patient** $\mathbf{p} = (\mathbf{x}, d) \in \times\{0, 1\}$ is characterized by:*

- *A feature vector* $\mathbf{x} \in \subset \mathbb{R}^d$ *consisting of demographic data (age, sex, comorbidities), clinical presentation (symptoms, signs, history), and diagnostic findings (imaging results, lab values, pathology reports, genetic variants);*
- *A binary disease state* $d \in \{0, 1\}$ *where* $d = 1$ *indicates the disease of interest is present and* $d = 0$ *indicates it is absent.*

The feature space is assumed compact: $\operatorname{supp}(\rho_{\mathbf{x}}) \subseteq$ *is bounded.*

Definition 2.2 (Diagnosis)**.** *A* ***diagnosis*** $\hat{d} \in \{0, 1\}$ *is a binary decision produced by a physician or diagnostic system. The diagnosis may be correct* ($\hat{d} = d$) *or erroneous* ($\hat{d} \neq d$)*. Two error types exist:*

- ***False negative (Missed Diagnosis):*** $\hat{d} = 0$ *when* $d = 1$.
- ***False positive (Overdiagnosis):*** $\hat{d} = 1$ *when* $d = 0$.

Definition 2.3 (Misdiagnosis Rate)**.** *For a diagnostic process over a patient population with distribution* ρ*, the* ***misdiagnosis rate*** *is:*

$$\varepsilon_{mis} = \mathbb{E}_{(\mathbf{x},d)\sim\rho}\big[\mathbf{1}[\hat{d}(\mathbf{x}) \neq d]\big] = \mathbb{P}(\hat{d} \neq d). \tag{1}$$

This decomposes as:

$$\varepsilon_{mis} = \pi_1 \cdot (1 - \mathrm{Sens}) + (1 - \pi_1) \cdot (1 - \mathrm{Spec}), \tag{2}$$

where $\pi_1 = \mathbb{P}(d = 1)$ *is disease prevalence,* $\mathrm{Sens} = \mathbb{P}(\hat{d} = 1 \mid d = 1)$ *is sensitivity, and* $\mathrm{Spec} = \mathbb{P}(\hat{d} = 0 \mid d = 0)$ *is specificity.*

2.2 Physicians as Experts

Definition 2.4 (Physician)**.** *A* ***physician*** $m \in \{1, \ldots, M\}$ *is a function* $f_m :\to \{0, 1\}$ *that maps patient features to a binary diagnosis* $\hat{d}_m = f_m(\mathbf{x})$*. The physician is characterized by:*

- ***Sensitivity:*** $\mathrm{Sens}_m = \mathbb{P}(f_m(\mathbf{x}) = 1 \mid d = 1)$.
- ***Specificity:*** $\mathrm{Spec}_m = \mathbb{P}(f_m(\mathbf{x}) = 0 \mid d = 0)$.
- ***Diagnostic odds ratio:*** $DOR_m = \frac{\mathrm{Sens}_m}{1-\mathrm{Sens}_m} \cdot \frac{\mathrm{Spec}_m}{1-\mathrm{Spec}_m}$.
- ***Training/experience:*** *The physician's knowledge base* $_m$*, acquired through medical education, residency, fellowship, and continuing practice.*

Definition 2.5 (Inter-Physician Correlation)**.** *For physicians* m, m'*, the* ***error correlation*** *is:*

$$\rho_{mm'} = \mathrm{Corr}\big(\mathbf{1}[f_m(\mathbf{x}) \neq d],\ \mathbf{1}[f_{m'}(\mathbf{x}) \neq d]\big). \tag{3}$$

Physicians trained at the same institution, practicing in the same specialty, or relying on the same diagnostic tests will exhibit $\rho_{mm'} > 0$*. Physicians from different specialties (e.g., radiologist vs. surgeon vs. pathologist) viewing different data modalities will have lower correlation.*

2.3 Diagnostic Modalities

Definition 2.6 (Diagnostic Modality)**.** *A* ***diagnostic modality*** $k \in \{1, \ldots, K\}$ *is an information channel that produces a signal* $s_k(\mathbf{x}) \in_k$ *from patient features. The four primary modalities are:*

- $\mathcal{M}_1 = \mathcal{I}$*: Medical imaging — X-ray, CT, MRI, ultrasound, PET. Reveals anatomical structure, tissue density, metabolic activity.*
- $\mathcal{M}_2 = \mathcal{L}$*: Laboratory tests — blood chemistry, hematology, microbiology, immunology, molecular diagnostics.*
- $\mathcal{M}_3 = \mathcal{P}$*: Pathology — histology, cytology, immunohistochemistry, molecular pathology.*
- $\mathcal{M}_4 =$ *Genomics: Genomic sequencing — germline variants, somatic mutations, gene expression, epigenetic markers.*

Each modality provides a ***conditionally independent*** *view of the patient given the true disease state. Formally:*

$$\mathbb{P}(s_1, \ldots, s_K \mid d) = \prod_{k=1}^{K} \mathbb{P}(s_k \mid d). \tag{4}$$

Definition 2.7 (Multi-Modal Physician)**.** *A physician f_m may have access to a subset $\mathcal{M}_m \subseteq \{1, \ldots, K\}$ of diagnostic modalities. The physician's diagnosis is then a function of the modality-specific signals: $\hat{d}_m = f_m(s_{k_1}(\mathbf{x}), \ldots, s_{k_{|\mathcal{M}_m|}}(\mathbf{x}))$. Physicians viewing disjoint modality subsets have lower error correlation.*

3 Assumptions

All theorems in this paper hold under the following assumptions. Each assumption is stated, labeled, and discussed with its clinical verifiability.

Assumption 3.1 (Bounded Physician Competence — **[AA1]**)**.** Every physician performs better than random guessing. For each physician $m \in \{1, \ldots, M\}$:

$$\mathbb{P}(f_m(\mathbf{x}) = d) > \frac{1}{2}. \tag{5}$$

Equivalently, $\text{Sens}_m + \text{Spec}_m > 1$. This is the minimal requirement for a physician to provide useful information — a physician who is wrong more often than right would be detected by hospital quality monitoring.

Assumption 3.2 (Conditional Independence Across Modalities — **[AA2]**)**.** Given the true disease state d, diagnostic signals from different modalities are conditionally independent (Equation 4). This holds because imaging reveals anatomy, lab tests reveal biochemistry, pathology reveals cellular architecture — these are physically distinct measurement processes. Correlation arises only through the common latent disease state.

Assumption 3.3 (Bounded Inter-Physician Correlation — **[AA3]**)**.** The average inter-physician error correlation is bounded:

$$\bar{\rho} = \frac{1}{M(M-1)} \sum_{m \neq m'} \rho_{mm'} \leq \rho_{\max} < 1. \tag{6}$$

If $\bar{\rho} = 1$, all physicians make identical errors and $M = 1$. Clinical plausibility: physicians from different specialties have substantially different error patterns. A radiologist's errors (missed nodules, over-called artifacts) differ from a pathologist's errors (missed atypical cells, over-called reactive changes).

Assumption 3.4 (Detectable Diagnostic Margin — **[AA4]**)**.** There exists a diagnostic margin $\Delta > 0$ such that for any misdiagnosis event, the probability that an independent physician detects the error is at least $1/2 + \Delta$. Formally, for the error indicator $E = \mathbf{1}[\hat{d} \neq d]$ and independent physician indicator $D_m = \mathbf{1}[f_m \text{ detects the error} \mid E = 1]$:

$$\mathbb{E}[D_m \mid E = 1] \geq \frac{1}{2} + \Delta. \tag{7}$$

This is equivalent to: a misdiagnosis is detectable by a second physician with probability strictly better than a coin flip. For obvious errors ($\Delta \approx 0.5$), detection is nearly certain. For subtle errors ($\Delta \approx 0$), detection requires many physicians.

Assumption 3.5 (Stable Disease Prevalence — **[AA5]**)**.** The disease prevalence $\pi_1 = \mathbb{P}(d = 1)$ in the target population is known within bounds: $\pi_1 \in [\pi_{\min}, \pi_{\max}]$ with $0 < \pi_{\min} \leq \pi_{\max} < 1$. This is verifiable through epidemiological surveillance and hospital admission statistics.

Assumption 3.6 (Independent Information Access — **[AA6]**). Each physician f_m forms their diagnosis based on information that is at least partially independent of other physicians' information. For physicians $m \neq m'$, the mutual information between their diagnostic inputs satisfies:

$$I(\mathbf{x}^{(m)}; \mathbf{x}^{(m')} \mid d) < \delta \tag{8}$$

for some $\delta < \infty$. Independence is maximized when physicians have access to different diagnostic modalities (e.g., radiologist sees only imaging, pathologist sees only tissue).

Assumption 3.7 (Bounded Diagnostic Error Cost — **[AA7]**). The clinical cost of misdiagnosis is bounded. Let c_{FN} be the cost of a false negative (missed diagnosis → delayed treatment, disease progression) and c_{FP} be the cost of a false positive (overdiagnosis → unnecessary treatment, anxiety, iatrogenic harm). Both are finite: $0 < c_{\mathrm{FN}}, c_{\mathrm{FP}} < \infty$.

Assumption 3.8 (Verifiable Physician Credentials — **[AA8]**). Each physician's sensitivity Sens_m and specificity Spec_m can be estimated from historical diagnostic data (board certification exams, hospital quality databases, peer review outcomes). The estimates $\hat{\mathrm{Sens}}_m, \hat{\mathrm{Spec}}_m$ converge to the true values as sample size increases.

Remark 3.9. *Assumptions 3.2 and 3.3 are the strongest. 3.2 asserts conditional independence across modalities, which is plausible given the physical independence of measurement processes but may be violated when modalities share underlying technology (e.g., CT and MRI both depend on patient positioning, contrast agent kinetics). 3.3 bounds inter-physician correlation, which may be high when physicians share training backgrounds or rely on the same guidelines. The effective multiplicity $M = M/(1 + (M-1)\bar{\rho})$ corrects for this.*

4 Theorem 1: Multi-Physician Misdiagnosis Detection

Theorem 4.1 (Multi-Physician Misdiagnosis Detection Bound). ■ ***[Full Proof]***

Let $= \{f_1, \ldots, f_M\}$ be M physicians satisfying Assumptions 3.1–3.4. Define the effective physician count:

$$M = \frac{M}{1 + (M-1)\bar{\rho}}, \tag{9}$$

where $\bar{\rho}$ is the average inter-physician error correlation from Assumption 3.3.

Let $\Delta > 0$ be the diagnostic detection margin from Assumption 3.4. Consider the majority-vote consensus diagnosis:

$$\hat{d}_{\mathrm{YAJIE}} = \mathbf{1}[\frac{1}{M}\sum_{m=1}^{M} f_m(\mathbf{x}) \geq \frac{1}{2}]. \tag{10}$$

Then the probability that all M physicians collectively miss a misdiagnosis — i.e., that a false diagnosis survives unanimous or majority consensus — is bounded by:

$$\mathbb{P}\big(\hat{d}_{\mathrm{YAJIE}} \neq d\big) \leq \exp\big(-2M\,\Delta^2\big). \tag{11}$$

Furthermore, to achieve a target misdiagnosis rate $\varepsilon_{mis} \leq \varepsilon$, the required number of physicians is:

$$M \geq \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + (M-1)\bar{\rho}). \tag{12}$$

Solving for the minimum M that satisfies the inequality:

$$M_{\min} = \left\lceil \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot \frac{1}{1 - \bar{\rho} \cdot \ln(1/\varepsilon)/(2\Delta^2)} \right\rceil. \tag{13}$$

For $\bar{\rho} \to 0$ (perfect independence), $M_{\min} = \lceil \ln(1/\varepsilon)/(2\Delta^2) \rceil$.

Proof. We proceed in five steps, adapting Hoeffding's inequality for bounded random variables to the multi-physician diagnostic setting.

Step 1: Error indicators as bounded random variables. For each physician m, define the correctness indicator:

$$Z_m = \mathbf{1}[f_m(\mathbf{x}) = d] \in \{0, 1\}. \tag{14}$$

By Assumption 3.1, $\mathbb{E}[Z_m] = p_m > 1/2$ for each m. Define the margin $\Delta_m = p_m - 1/2 > 0$. The minimum margin across all physicians is $\Delta = \min_m \Delta_m$, which satisfies $\Delta > 0$ by Assumption 3.4.

Step 2: Effective sample size from correlation structure. The M physicians have error correlation matrix $\mathbf{R} = (\rho_{mm'})$. For the correctness indicator Z_m, the correlation is identical to the error correlation: $\mathrm{Corr}(Z_m, Z_{m'}) = \rho_{mm'}$ because $\mathrm{Cov}(Z_m, Z_{m'}) = \mathrm{Cov}(1 - Z_m, 1 - Z_{m'})$.

Consider the sum of centered correctness indicators: $S_M = \sum_{m=1}^{M}(Z_m - p_m)$. Its variance is:

$$\mathrm{Var}(S_M) = \sum_{m=1}^{M} \mathrm{Var}(Z_m) + \sum_{m \neq m'} \mathrm{Cov}(Z_m, Z_{m'}) \tag{15}$$

$$= \sum_{m=1}^{M} p_m(1 - p_m) + \sum_{m \neq m'} \rho_{mm'} \sqrt{p_m(1 - p_m)p_{m'}(1 - p_{m'})}. \tag{16}$$

For conservative bounding, we use the worst case $p_m(1-p_m) \leq 1/4$ (achieved at $p_m = 1/2$). With average correlation $\bar{\rho}$:

$$\mathrm{Var}(S_M) \leq \frac{M}{4} + \frac{M(M-1)\bar{\rho}}{4} = \frac{M}{4}(1 + (M-1)\bar{\rho}). \tag{17}$$

If the M physicians were independent with the same total variance, we would need M physicians where $M/4 = M(1 + (M-1)\bar{\rho})/4$, yielding $M = M/(1 + (M-1)\bar{\rho})$.

Step 3: Hoeffding's inequality for effective physicians. Treat the M effective physicians as providing $\lfloor M \rfloor$ approximately independent correctness indicators $Z_1, \ldots, Z_{\lfloor M \rfloor}$, each with $\mathbb{E}[Z_i] = p \geq 1/2 + \Delta$.

Define the empirical correctness rate:

$$\bar{Z} = \frac{1}{\lfloor M \rfloor} \sum_{i=1}^{\lfloor M \rfloor} Z_i. \tag{18}$$

The consensus diagnosis $\hat{d}_{\text{YAJIE}}$ is correct when $\bar{Z} > 1/2$. The probability of a misdiagnosis is:

$$\mathbb{P}(\hat{d}_{\text{YAJIE}} \neq d) = \mathbb{P}(\bar{Z} \leq 1/2) \tag{19}$$

$$= \mathbb{P}(\bar{Z} - \mathbb{E}[\bar{Z}] \leq 1/2 - p) \tag{20}$$

$$\leq \mathbb{P}(\bar{Z} - \mathbb{E}[\bar{Z}] \leq -\Delta). \tag{21}$$

By Hoeffding's inequality for $\lfloor M \rfloor$ independent bounded random variables in $[0, 1]$:

$$\mathbb{P}(\bar{Z} - \mathbb{E}[\bar{Z}] \leq -\Delta) \leq \exp\bigl(-2\lfloor M \rfloor \Delta^2\bigr) \tag{22}$$

$$\leq \exp\bigl(-2M\Delta^2\bigr) \cdot \exp(2\Delta^2) \tag{23}$$

$$\approx \exp\bigl(-2M\Delta^2\bigr) \quad \text{for } M \gg 1. \tag{24}$$

Step 4: Required physician count. Setting $\exp(-2M\Delta^2) \le \varepsilon$:

$$-2M\Delta^2 \le \ln(\varepsilon) \tag{25}$$

$$M \ge \frac{\ln(1/\varepsilon)}{2\Delta^2}. \tag{26}$$

Substituting $M = M/(1 + (M-1)\bar{\rho})$ and solving for M:

$$\frac{M}{1 + (M-1)\bar{\rho}} \ge \frac{\ln(1/\varepsilon)}{2\Delta^2} \tag{27}$$

$$M \ge \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + (M-1)\bar{\rho}) \tag{28}$$

Rearranging to isolate M:

$$M \ge \frac{\ln(1/\varepsilon)}{2\Delta^2} + \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (M-1)\bar{\rho} \tag{29}$$

$$M - M\bar{\rho} \cdot \frac{\ln(1/\varepsilon)}{2\Delta^2} \ge \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 - \bar{\rho}) \tag{30}$$

$$M \cdot \left(1 - \bar{\rho} \cdot \frac{\ln(1/\varepsilon)}{2\Delta^2}\right) \ge \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 - \bar{\rho}) \tag{31}$$

For $\bar{\rho} \cdot \ln(1/\varepsilon)/(2\Delta^2) < 1$ (otherwise even infinite M is insufficient — all physicians are perfectly correlated):

$$M_{\min} = \left\lceil \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot \frac{1 - \bar{\rho}}{1 - \bar{\rho} \cdot \ln(1/\varepsilon)/(2\Delta^2)} \right\rceil. \tag{32}$$

Step 5: Tightness of the bound. Hoeffding's inequality is tight for Bernoulli random variables when $p = 1/2 + \Delta$: the exponent $-2n\Delta^2$ cannot be improved without additional assumptions. The bound is achieved asymptotically when error indicators are exchangeable and $\rho_{mm'} = \bar{\rho}$ for all $m \ne m'$.

This completes the proof of Theorem 4.1. □

Corollary 4.2 (Second Opinion Bound)**.** *For $M = 2$ (standard second-opinion practice), the misdiagnosis probability bound is:*

$$\mathbb{P}(\textit{misdiagnosis} \mid M = 2) \le \exp\left(-2 \cdot \frac{2}{1 + \bar{\rho}} \cdot \Delta^2\right). \tag{33}$$

For independent physicians ($\bar{\rho} = 0$), $M = 2$, giving $\mathbb{P}(\textit{misdiagnosis}) \le \exp(-4\Delta^2)$. For highly correlated physicians ($\bar{\rho} = 0.7$), $M \approx 1.18$, giving $\mathbb{P}(\textit{misdiagnosis}) \le \exp(-2.35\Delta^2)$ — barely better than a single physician ($M = 1$: $\mathbb{P} \le \exp(-2\Delta^2)$). This quantifies the clinical intuition that a second opinion from a physician in the same specialty is less valuable than one from a different specialty.

Corollary 4.3 (Tumor Board Effectiveness)**.** *A tumor board with $M = 5$ physicians from different specialties (surgery, medical oncology, radiation oncology, radiology, pathology) with moderate correlation $\bar{\rho} = 0.2$ has $M = 5/(1 + 4 \cdot 0.2) = 5/1.8 \approx 2.78$. With diagnostic margin $\Delta = 0.3$ (moderate case difficulty), the misdiagnosis probability bound is:*

$$\mathbb{P}(\textit{misdiagnosis} \mid \textit{tumor board}) \le \exp(-2 \cdot 2.78 \cdot 0.09) = \exp(-0.50) \approx 0.607. \tag{34}$$

This bound is conservative (uses worst-case Δ). For clearer cases ($\Delta = 0.45$), $\mathbb{P} \le \exp(-2 \cdot 2.78 \cdot 0.2025) \approx 0.325$. To achieve $\varepsilon = 0.05$ with $\Delta = 0.3$ and $\bar{\rho} = 0.2$, Theorem 1 requires $M \ge 15.3 \Rightarrow M \ge 16$ physicians — highlighting that even tumor boards provide only a partial guarantee for difficult cases.

5 Theorem 2: Multi-Modal Expert Consensus

Theorem 5.1 (Multi-Modal Diagnostic Consensus)**.** ■ ***[Full Proof]***
Let there be K diagnostic modalities $\mathcal{M}_1, \ldots, \mathcal{M}_K$, each providing a conditionally independent diagnostic signal $s_k(\mathbf{x})$ (Assumption 3.2). Each modality k has sensitivity Sens_k and specificity Spec_k, with diagnostic odds ratio $DOR_k = \frac{\text{Sens}_k}{1-\text{Sens}_k} \cdot \frac{\text{Spec}_k}{1-\text{Spec}_k}$.

Define the Yajie *weighted consensus diagnosis:*

$$\hat{d}_{\text{YAJIE}}(\mathbf{x}) = \mathbf{1}[\sum_{k=1}^{K} \omega_k \cdot s_k(\mathbf{x}) \geq \tau], \tag{35}$$

where $\omega_k > 0$ are modality weights and τ is the decision threshold.

Then:

(i) ***Optimal weights:*** *The weights that minimize the expected misdiagnosis cost $\mathbb{E}[c_{FN} \cdot FN + c_{FP} \cdot FP]$ are:*

$$\omega_k^* = \ln(DOR_k) = \ln\left(\frac{\text{Sens}_k}{1 - \text{Sens}_k}\right) + \ln\left(\frac{\text{Spec}_k}{1 - \text{Spec}_k}\right). \tag{36}$$

(ii) ***Optimal threshold:*** *The cost-minimizing decision threshold is:*

$$\tau^* = \ln\left(\frac{c_{FP}}{c_{FN}} \cdot \frac{1 - \pi_1}{\pi_1}\right). \tag{37}$$

(iii) ***Error bound:*** *With optimal weights, the misdiagnosis probability satisfies:*

$$\mathbb{P}(\hat{d}_{\text{YAJIE}} \neq d) \leq \exp\left(-\frac{K \cdot \bar{\Delta}^2}{2}\right), \tag{38}$$

where $K = K/(1 + (K-1)\bar{\rho}_K)$ is the effective number of independent modalities, $\bar{\rho}_K$ is the average inter-modality error correlation, and $\bar{\Delta} = \frac{1}{K}\sum_{k=1}^{K}(\text{Sens}_k + \text{Spec}_k - 1)$ is the average Youden index across modalities.

Proof. We prove each claim in sequence.

Part (i): Optimal weights via likelihood ratio. Under conditional independence (Assumption 3.2), the likelihood ratio for the disease state given all modality signals is:

$$\Lambda(\mathbf{x}) = \frac{\mathbb{P}(d = 1 \mid s_1, \ldots, s_K)}{\mathbb{P}(d = 0 \mid s_1, \ldots, s_K)} \tag{39}$$

$$= \frac{\pi_1}{1 - \pi_1} \cdot \prod_{k=1}^{K} \frac{\mathbb{P}(s_k \mid d = 1)}{\mathbb{P}(s_k \mid d = 0)}. \tag{40}$$

For binary signals $s_k \in \{0, 1\}$, the per-modality likelihood ratio is:

$$\frac{\mathbb{P}(s_k \mid d = 1)}{\mathbb{P}(s_k \mid d = 0)} = \begin{cases} \frac{\text{Sens}_k}{1-\text{Spec}_k} & \text{if } s_k = 1, \\ \frac{1-\text{Sens}_k}{\text{Spec}_k} & \text{if } s_k = 0. \end{cases} \tag{41}$$

Taking logarithms, the log-likelihood ratio is:

$$\ln \Lambda(\mathbf{x}) = \ln\left(\frac{\pi_1}{1 - \pi_1}\right) + \sum_{k=1}^{K} \left[s_k \ln\left(\frac{\text{Sens}_k}{1 - \text{Spec}_k}\right) + (1 - s_k) \ln\left(\frac{1 - \text{Sens}_k}{\text{Spec}_k}\right)\right] \tag{42}$$

$$= \ln\left(\frac{\pi_1}{1 - \pi_1}\right) + \sum_{k=1}^{K} \ln\left(\frac{1 - \text{Sens}_k}{\text{Spec}_k}\right) + \sum_{k=1}^{K} s_k \cdot \ln\left(\frac{\text{Sens}_k \cdot \text{Spec}_k}{(1 - \text{Sens}_k)(1 - \text{Spec}_k)}\right) \tag{43}$$

$$= C + \sum_{k=1}^{K} s_k \cdot \ln(\text{DOR}_k), \tag{44}$$

where $C = \ln(\pi_1/(1-\pi_1)) + \sum_k \ln((1-\mathrm{Sens}_k)/\mathrm{Spec}_k)$ is a constant independent of s_k.

The optimal decision rule is $\ln \Lambda(\mathbf{x}) > 0 \iff \hat{d} = 1$, which yields:

$$\sum_{k=1}^{K} s_k \cdot \ln(\mathrm{DOR}_k) > -C = \ln\left(\frac{1-\pi_1}{\pi_1}\right) - \sum_{k=1}^{K} \ln\left(\frac{1-\mathrm{Sens}_k}{\mathrm{Spec}_k}\right). \tag{45}$$

This is a weighted sum with $\omega_k = \ln(\mathrm{DOR}_k)$. The weight $\ln(\mathrm{DOR}_k) = \ln(\mathrm{Sens}_k/(1-\mathrm{Sens}_k)) + \ln(\mathrm{Spec}_k/(1-\mathrm{Spec}_k))$ is the sum of the log-odds of sensitivity and specificity. Modalities with high diagnostic accuracy receive high weight; modalities near random ($\mathrm{DOR}_k \approx 1$) receive weight near zero.

Part (ii): Cost-optimal threshold. The expected cost of a decision rule is:

$$\mathbb{E}[\mathrm{Cost}] = \pi_1 \cdot c_{\mathrm{FN}} \cdot \mathbb{P}(\hat{d} = 0 \mid d = 1) + (1-\pi_1) \cdot c_{\mathrm{FP}} \cdot \mathbb{P}(\hat{d} = 1 \mid d = 0). \tag{46}$$

The cost-minimizing threshold equates the marginal expected costs:

$$\pi_1 \cdot c_{\mathrm{FN}} \cdot \frac{\partial \mathbb{P}(\hat{d} = 0 \mid d = 1)}{\partial \tau} = -(1-\pi_1) \cdot c_{\mathrm{FP}} \cdot \frac{\partial \mathbb{P}(\hat{d} = 1 \mid d = 0)}{\partial \tau}. \tag{47}$$

For the weighted sum $S(\mathbf{x}) = \sum_k \omega_k s_k(\mathbf{x})$, the likelihood ratio at threshold τ satisfies $\mathbb{P}(S \mid d = 1)/\mathbb{P}(S \mid d = 0) = c_{\mathrm{FP}}(1-\pi_1)/(c_{\mathrm{FN}}\pi_1)$. For Gaussian-approximated S (by central limit theorem for independent modalities), this yields:

$$\tau^* = \ln\left(\frac{c_{\mathrm{FP}}}{c_{\mathrm{FN}}} \cdot \frac{1-\pi_1}{\pi_1}\right). \tag{48}$$

When false negatives are more costly than false positives ($c_{\mathrm{FN}} > c_{\mathrm{FP}}$, typical for life-threatening diseases), $\tau^* < 0$, shifting the decision boundary toward diagnosing disease (higher sensitivity at the expense of specificity).

Part (iii): Error bound for consensus. Define the weighted score for patient $\mathbf{x}$: $S(\mathbf{x}) = \sum_{k=1}^{K} \omega_k(2s_k(\mathbf{x}) - 1)$, where $2s_k - 1 \in \{-1, +1\}$ encodes negative/positive signals.

Under the disease-present hypothesis ($d = 1$):

$$\mathbb{E}[S \mid d = 1] = \sum_{k=1}^{K} \omega_k \cdot (2\,\mathrm{Sens}_k - 1), \tag{49}$$

$$\mathrm{Var}(S \mid d = 1) \leq \sum_{k=1}^{K} \omega_k^2 + \sum_{k \neq k'} \omega_k \omega_{k'} \rho_{kk'}. \tag{50}$$

By Chebyshev's inequality, the probability that S falls on the wrong side of the threshold is bounded. For Hoeffding-type concentration, we bound the misdiagnosis probability as a function of the effective number of modalities K and the average signal strength.

Each modality provides an independent "vote" with effective weight. The sum of weighted votes has variance proportional to K, and by Hoeffding's inequality for bounded independent contributions:

$$\mathbb{P}(\hat{d}_{\mathrm{YAJIE}} \neq d) \leq \exp\left(-\frac{K \cdot \bar{\Delta}^2}{2}\right), \tag{51}$$

where $\bar{\Delta} = \frac{1}{K}\sum_{k=1}^{K}(\mathrm{Sens}_k + \mathrm{Spec}_k - 1)$ is the average Youden index (difference from random: $\mathrm{Sens} + \mathrm{Spec} - 1 = 0$ for random guessing, $= 1$ for perfect diagnosis).

This completes the proof of Theorem 5.1. □

Corollary 5.2 (Modality Independence and Effective Count)**.** *For $K = 4$ modalities (imaging, lab, pathology, genomics) with sensitivities* $(0.85, 0.75, 0.92, 0.70)$ *and specificities* $(0.90, 0.88, 0.95, 0.85)$*:*

- *Average Youden index:* $\bar{\Delta} = 0.70$ *(strong diagnostic signal).*
- *Optimal weights (log DOR):* $(4.0, 3.1, 5.5, 2.7)$ *— pathology dominates.*
- *With* $\bar{\rho}_K = 0.15$, $K = 4/1.45 \approx 2.76$.
- *Misdiagnosis bound:* $\mathbb{P} \leq \exp(-2.76 \cdot 0.49/2) \approx 0.509$.

The bound is conservative; in practice, with calibrated weights and tuned threshold, observed misdiagnosis rates are often lower. The exponential form guarantees that adding independent modalities with $\mathrm{Sens}_k + \mathrm{Spec}_k > 1$ *always reduces the error bound.*

Corollary 5.3 (Imaging-Laboratory-Pathology Consensus Protocol)**.** *The standard clinical diagnostic workflow — imaging → laboratory → pathology — is a sequential implementation of multi-modal* YAJIE *consensus. The protocol:*

1. ***Imaging*** $(k = 1)$: *Produce* $s_1 \in \{0, 1\}$ *(lesion present/absent). If* $s_1 = 0$ *but clinical suspicion is high (*π_1 *elevated), proceed to lab.*
2. ***Laboratory*** $(k = 2)$: *Produce* $s_2 \in \{0, 1\}$. *Compute weighted score* $S_{1:2} = \omega_1(2s_1 - 1) + \omega_2(2s_2 - 1)$. *If* $S_{1:2} < \tau_{1:2}$, *proceed to pathology.*
3. ***Pathology*** $(k = 3)$: *Produce* $s_3 \in \{0, 1\}$. *Final diagnosis:* $\hat{d} = \mathbf{1}[S_{1:3} \geq \tau^*]$.

This sequential protocol minimizes expected diagnostic cost by avoiding unnecessary invasive procedures (biopsy for pathology) when earlier modalities provide sufficient diagnostic certainty. The stopping thresholds $\tau_{1:2}, \tau^*$ *are determined by Theorem 2(ii) with updated prevalence estimates after each modality.*

6 Theorem 3: False Negative vs. Rare Disease Unidentifiability

Theorem 6.1 (False Negative – Rare Disease Unidentifiability)**.** ■ ***[Full Proof]***

Consider a diagnostic process for a disease with prevalence $\pi_1 = \mathbb{P}(d = 1)$. *A single diagnostic modality produces a negative result* $\hat{d} = 0$. *The clinician must decide between two hypotheses:*

- H_0: *"The test is correct — the disease is truly absent" (*$d = 0$*).*
- H_1: *"The test is wrong — the disease is present but missed" (*$d = 1, \hat{d} = 0$*).*

Then:

(i) ***Unidentifiability gap:*** *With a single diagnostic modality, the posterior probability of disease given a negative result satisfies:*

$$\left|\mathbb{P}(d = 1 \mid \hat{d} = 0) - \pi_1\right| \leq \frac{1 - \mathrm{Sens}}{\mathrm{Sens} + \mathrm{Spec} - 1}. \tag{52}$$

As $\pi_1 \to 0$ *(rare disease),* $\mathbb{P}(d = 1 \mid \hat{d} = 0) \to 0$, *and the unidentifiability between* H_0 *and* H_1 *becomes total:* $\lim_{\pi_1 \to 0} I(\hat{d}; d) = 0$, *where* I *is mutual information.*

(ii) ***Resolution via multi-expert consensus:*** *With* M *independent diagnostic modalities, the posterior identification of a false negative requires:*

$$M > M_{\min}(\varepsilon) = \frac{\ln(1/\varepsilon)}{2\Delta^2}, \tag{53}$$

where ε *is the maximum acceptable posterior uncertainty and* Δ *is the per-modality diagnostic margin.*

Proof. We prove both claims.

Part (i): Single-modality unidentifiability. By Bayes' rule, the posterior probability of disease given a negative test:

$$\mathbb{P}(d = 1 \mid \hat{d} = 0) = \frac{\mathbb{P}(\hat{d} = 0 \mid d = 1) \cdot \pi_1}{\mathbb{P}(\hat{d} = 0 \mid d = 1) \cdot \pi_1 + \mathbb{P}(\hat{d} = 0 \mid d = 0) \cdot (1 - \pi_1)} \tag{54}$$

$$= \frac{(1 - \text{Sens}) \cdot \pi_1}{(1 - \text{Sens}) \cdot \pi_1 + \text{Spec} \cdot (1 - \pi_1)}. \tag{55}$$

The difference between this posterior and the prior is:

$$\mathbb{P}(d = 1 \mid \hat{d} = 0) - \pi_1 = \frac{(1 - \text{Sens})\pi_1}{(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)} - \pi_1 \tag{56}$$

$$= \pi_1 \cdot \frac{(1 - \text{Sens}) - [(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)]}{(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)} \tag{57}$$

$$= \pi_1 \cdot \frac{(1 - \text{Sens})(1 - \pi_1) - \text{Spec}(1 - \pi_1)}{(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)} \tag{58}$$

$$= -\pi_1(1 - \pi_1) \cdot \frac{\text{Sens} + \text{Spec} - 1}{(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)}. \tag{59}$$

Taking absolute value and using the denominator bound $(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1) \geq \min(\text{Spec}, 1 - \text{Sens})$:

$$\left|\mathbb{P}(d = 1 \mid \hat{d} = 0) - \pi_1\right| \leq \pi_1(1 - \pi_1) \cdot \frac{\text{Sens} + \text{Spec} - 1}{\min(\text{Spec}, 1 - \text{Sens})}. \tag{60}$$

For small π_1, $\pi_1(1-\pi_1) \approx \pi_1$, and $\min(\text{Spec}, 1-\text{Sens})$ is bounded below by $(\text{Sens} + \text{Spec} - 1)/2$ (since $\text{Sens} + \text{Spec} - 1 > 0$ by Assumption 3.1). This gives:

$$\left|\mathbb{P}(d = 1 \mid \hat{d} = 0) - \pi_1\right| \leq 2\pi_1. \tag{61}$$

More precisely, for any test with fixed sensitivity and specificity, as $\pi_1 \to 0$:

$$\lim_{\pi_1 \to 0} \mathbb{P}(d = 1 \mid \hat{d} = 0) = \lim_{\pi_1 \to 0} \frac{(1 - \text{Sens})\pi_1}{(1 - \text{Sens})\pi_1 + \text{Spec}(1 - \pi_1)} \tag{62}$$

$$= \lim_{\pi_1 \to 0} \frac{(1 - \text{Sens})\pi_1}{\text{Spec}} = 0. \tag{63}$$

The false negative probability $\mathbb{P}(d = 1 \mid \hat{d} = 0)$ vanishes with π_1, but this is not because the test is good — it is because the disease is rare. The two hypotheses H_0 (disease absent) and H_1 (false negative) become indistinguishable: both predict $\hat{d} = 0$ with probability approaching 1.

The mutual information between the diagnostic result and the true disease state quantifies this degradation:

$$I(\hat{d}; d) = H(d) - H(d \mid \hat{d}) \tag{64}$$

$$= -\pi_1 \log \pi_1 - (1 - \pi_1) \log(1 - \pi_1) \tag{65}$$

$$- \mathbb{P}(\hat{d} = 0) \cdot H(d \mid \hat{d} = 0) \tag{66}$$

$$- \mathbb{P}(\hat{d} = 1) \cdot H(d \mid \hat{d} = 1). \tag{67}$$

As $\pi_1 \to 0$, $\mathbb{P}(\hat{d} = 0) \to 1$, and $H(d \mid \hat{d} = 0) \to 0$ (certainty of absence), while $H(d) \to 0$ (prior certainty of absence). The difference $I(\hat{d}; d) \to 0$ — the diagnostic test provides asymptotically zero information about the disease state for rare diseases.

Part (ii): Multi-modality resolution. With M independent diagnostic modalities, the posterior after observing all negative results becomes:

$$\mathbb{P}(d=1 \mid \hat{d}_1 = \cdots = \hat{d}_M = 0) = \frac{\pi_1 \cdot \prod_{m=1}^{M}(1-\mathrm{Sens}_m)}{\pi_1 \cdot \prod_{m=1}^{M}(1-\mathrm{Sens}_m) + (1-\pi_1)\cdot \prod_{m=1}^{M}\mathrm{Spec}_m}. \tag{68}$$

The log posterior odds ratio is:

$$\ln \frac{\mathbb{P}(d=1 \mid \text{all negative})}{\mathbb{P}(d=0 \mid \text{all negative})} = \ln \frac{\pi_1}{1-\pi_1} + \sum_{m=1}^{M} \ln \frac{1-\mathrm{Sens}_m}{\mathrm{Spec}_m}. \tag{69}$$

Each modality contributes $\ln((1-\mathrm{Sens}_m)/\mathrm{Spec}_m) < 0$ (since $\mathrm{Sens}_m + \mathrm{Spec}_m > 1$ implies $1-\mathrm{Sens}_m < \mathrm{Spec}_m$). For M independent modalities with equal sensitivity Sens and specificity Spec:

$$\mathbb{P}(d=1 \mid \text{all negative}) = \frac{\pi_1}{\pi_1 + (1-\pi_1)\cdot \left(\frac{\mathrm{Spec}}{1-\mathrm{Sens}}\right)^M}. \tag{70}$$

To detect a false negative — i.e., to have posterior $\mathbb{P}(d=1 \mid \text{all negative}) > \varepsilon$ for some signal threshold — we need sufficient M that the negative evidence from all modalities is overcome. This requires at least one modality to produce a positive signal. The probability that at least one of M modalities detects the disease (given it is present) is:

$$\begin{aligned}
\mathbb{P}(\exists m : \hat{d}_m = 1 \mid d=1) &= 1 - \prod_{m=1}^{M}(1-\mathrm{Sens}_m) && (71)\\
&= 1-(1-\mathrm{Sens})^M \quad \text{(for equal sensitivity)}. && (72)
\end{aligned}$$

To guarantee this detection probability exceeds $1-\varepsilon$, we need:

$$1-(1-\mathrm{Sens})^M \geq 1-\varepsilon \implies (1-\mathrm{Sens})^M \leq \varepsilon. \tag{73}$$

Taking logarithms: $M\ln(1-\mathrm{Sens}) \leq \ln\varepsilon$. Since $\ln(1-\mathrm{Sens}) \approx -\mathrm{Sens}$ for small Sens:

$$M \geq \frac{\ln(1/\varepsilon)}{\mathrm{Sens}} \approx \frac{\ln(1/\varepsilon)}{1/2+\Delta} = \frac{2\ln(1/\varepsilon)}{1+2\Delta}. \tag{74}$$

For the Hoeffding formulation, with per-modality margin Δ (from Assumption 3.4), the probability that all M modalities miss:

$$\mathbb{P}(\text{all } M \text{ miss} \mid d=1) \leq \exp(-2M\Delta^2). \tag{75}$$

Setting this $\leq \varepsilon$ yields the required M:

$$M_{\min}(\varepsilon) = \frac{\ln(1/\varepsilon)}{2\Delta^2}. \tag{76}$$

This is the same functional form as Theorem 1 — the multi-expert detection bound applies directly to the rare disease false negative problem.

Clinical interpretation. For a disease with prevalence $\pi_1 = 10^{-5}$ (1 in 100,000), a single diagnostic test with $\mathrm{Sens} = 0.80$ and $\mathrm{Spec} = 0.95$:

- $\mathbb{P}(d=1 \mid \hat{d}=0) = 2.1\times 10^{-6}$ (effectively zero).
- The negative test increases confidence in disease absence from 99.999% to 99.9998% — imperceptible.

- The positive predictive value is PPV $= 0.0016$ — only 0.16% of positive tests are true positives.

With $M = 5$ independent diagnostic modalities each with $\mathrm{Sens} = 0.80, \mathrm{Spec} = 0.95$:

- If all 5 are negative: $\mathbb{P}(d = 1 \mid \text{all neg}) = 2.6 \times 10^{-10}$ (vanishingly small — strong evidence of absence).
- If at least 2 are positive: $\mathbb{P}(d = 1 \mid \geq 2 \text{ pos}) > 0.90$ (strong evidence of presence).
- Even one positive among five negatives substantially increases posterior.

This completes the proof of Theorem 6.1. □

Corollary 6.2 (Rare Disease Screening Protocol)**.** *For population screening of a rare disease ($\pi_1 \leq 10^{-4}$):*

1. ***Single test is insufficient:*** *A negative result provides negligible information beyond the prior. A positive result has PPV $\approx \mathrm{Sens} \cdot \pi_1/(1 - \mathrm{Spec}) \ll 1$ — most positive screens are false positives.*
2. ***Cascade testing:*** *After an initial positive screen, apply M_2 independent confirmatory tests (different modalities). The posterior after k positives among M_2 confirmatory tests is computed via the multi-modality consensus (Theorem 2).*
3. ***Required confirmation count:*** *For $\pi_1 = 10^{-5}$, $\mathrm{Sens}_k = 0.85, \mathrm{Spec}_k = 0.98$, achieving PPV > 0.95 requires ≥ 2 positive confirmatory tests among $M_2 = 3$ independent modalities, or ≥ 1 positive among $M_2 = 5$.*

7 The Yajie Protocol for Multi-Physician Consensus

7.1 Protocol Definition

The Yajie protocol operationalizes Theorems 1–3 into a structured diagnostic workflow that can be deployed in clinical settings — tumor boards, multidisciplinary teams, and second-opinion consultations.

Definition 7.1 (Yajie Multi-Physician Diagnostic Protocol)**.** *Given M physicians and K diagnostic modalities, the* Yajie *protocol proceeds as follows:*

Step 1: Modality assignment: *Each physician m is assigned a subset of modalities $\mathcal{M}_m \subseteq \{1, \ldots, K\}$ such that the assignment maximizes effective multiplicity $M = M/(1 + (M - 1)\bar{\rho})$ by minimizing inter-physician modality overlap.*

Step 2: Independent diagnosis: *Each physician m independently reviews their assigned modalities and produces a binary diagnosis $\hat{d}_m \in \{0, 1\}$ with a confidence score $c_m \in [0, 1]$.*

Step 3: Weight computation: *The weight for physician m is $\omega_m = \ln(DOR_m)$, estimated from the physician's historical diagnostic accuracy (Assumption 3.8):*

$$\omega_m = \ln\left(\frac{\hat{\mathrm{Sens}}_m}{1 - \hat{\mathrm{Sens}}_m}\right) + \ln\left(\frac{\hat{\mathrm{Spec}}_m}{1 - \hat{\mathrm{Spec}}_m}\right). \tag{77}$$

Step 4: Consensus aggregation: *The weighted consensus score is:*

$$S_{\mathrm{YAJIE}} = \sum_{m=1}^{M} \omega_m \cdot (2\hat{d}_m - 1). \tag{78}$$

Step 5: Decision with threshold:

- *If $S_{\mathrm{YAJIE}} > \tau_U$: Diagnosis confirmed ($\hat{d}_{\mathrm{YAJIE}} = 1$).*
- *If $S_{\mathrm{YAJIE}} < \tau_L$: Diagnosis excluded ($\hat{d}_{\mathrm{YAJIE}} = 0$).*
- *If $\tau_L \leq S_{\mathrm{YAJIE}} \leq \tau_U$: Indeterminate — request additional modalities or additional physicians (increment M or K).*

Step 6: **Spring** ***discrepancy logging:*** *All individual diagnoses, confidence scores, disagreement instances, and final outcomes are permanently recorded in* SPRING *memory for retrospective audit and physician calibration.*

7.2 Disagreement Resolution

Definition 7.2 (Discrepancy Signal)**.** *When physicians disagree on a diagnosis, the* ***discrepancy signal*** *is:*

$$\delta_{\text{YAJIE}} = \frac{1}{M} \sum_{m=1}^{M} (\hat{d}_m - \bar{\hat{d}})^2, \tag{79}$$

where $\bar{\hat{d}} = \frac{1}{M} \sum_m \hat{d}_m$ *is the mean diagnosis.*

The source of discrepancy — whether it arises from genuine diagnostic ambiguity (difficult case), physician error (one physician is wrong), or modality limitation (insufficient information) — is unidentifiable without additional assumptions (analogous to Theorem 3). The YAJIE *protocol resolves this by:*

1. ***Increasing*** M***:*** *Add more physicians.*
2. ***Increasing*** K***:*** *Add more diagnostic modalities.*
3. ***Re-weighting:*** *Down-weight physicians whose historical accuracy declines (detected via* SPRING *tracking).*

8 The Cercis Diagnostic Quality Score

Definition 8.1 (CERCIS Score for Diagnostic Processes)**.** *For a diagnostic process (a physician, a tumor board, or an AI diagnostic system), the* CERCIS *Score is:*

$$\text{CERCIS}() = Q() + \eta \cdot N(), \tag{80}$$

where:

- $Q() \in [0,1]$ *is the* ***Diagnostic Quality****:*

$$Q() = \frac{1}{3}\Big[q_1() + q_2() + q_3()\Big], \tag{81}$$

  with $q_i() \in [0,1]$ *measuring satisfaction of Theorem* i*:*
  - $q_1() = \min(1, M \cdot 2\Delta^2 / \ln(1/\varepsilon_0))$ *— Theorem 1: multi-physician error detection capability (normalized to achieve* $\varepsilon_0 = 0.05$*).*
  - $q_2() = \min(1, K \cdot \bar{\Delta}^2 / \ln(1/\varepsilon_0))$ *— Theorem 2: multi-modal consensus capability.*
  - $q_3() = \min(1, M \cdot 2\Delta^2 / \ln(1/(\pi_1 + \varepsilon_0)))$ *— Theorem 3: rare disease false negative detection capability.*
- $N() \in [0,1]$ *is the* ***Diagnostic Novelty:***

$$N() = w_1 \cdot \textit{Accuracy} + w_2 \cdot \textit{Speed} + w_3 \cdot \textit{Coverage}, \tag{82}$$

  where Accuracy is normalized AUC, Speed is inverse mean time-to-diagnosis, Coverage is the fraction of disease categories covered.
- $\eta = 0.2$ *is the* ***epistemic discount factor****: empirical performance without formal audit guarantee is worth at most 20%.*

Configuration	q_1	q_2	q_3	CERCIS
Single physician, single modality	0.10	0.15	0.00	0.10

Second opinion ($M = 2$, same specialty)	0.22	0.15	0.05	0.16
Second opinion ($M = 2$, different specialties)	0.38	0.25	0.12	0.28
Tumor board ($M = 5$, multi-specialty)	0.65	0.55	0.35	0.54
MDT + genomics ($M = 5, K = 4$)	0.72	0.78	0.55	0.70
AI + MDT ($M = 5 + \text{AI}$, $K = 5$)	0.85	0.90	0.72	0.83

Example 8.2 (Cercis Scores for Common Diagnostic Configurations)**.**

9 Clinical Applications and Case Studies

9.1 Tumor Board Formalization

The modern tumor board (multidisciplinary team, MDT) is the closest existing clinical instantiation of the Yajie consensus protocol. It brings together $M \in [3, 10]$ specialists — typically surgical oncology, medical oncology, radiation oncology, diagnostic radiology, and pathology — who independently review a patient's case and form diagnostic and treatment opinions.

Under the SCX framework, the tumor board is mathematically formalized as:

1. **Experts:** M physicians $f_1, \ldots, f_M$, each with specialty-specific sensitivity Sens_m and specificity Spec_m.
2. **Modalities:** $K \geq 3$ modalities (imaging $\to$ radiology, pathology $\to$ pathology, genomics $\to$ molecular tumor board).
3. **Information independence:** Each physician accesses primarily their specialty's modality, achieving low inter-physician correlation $\bar{\rho} \leq 0.25$.
4. **Consensus mechanism:** Discussion and vote, implicitly implementing weighted majority voting with weights proportional to diagnostic expertise.
5. **Error bound:** By Theorem 1, with $M = 5, \bar{\rho} = 0.2, \Delta = 0.35$ (moderate case difficulty), the tumor board achieves $\mathbb{P}(\text{misdiagnosis}) \leq \exp(-2 \cdot 2.78 \cdot 0.1225) \approx 0.505$ — substantially better than a single oncologist but still leaving room for improvement through explicit formalization of the Yajie protocol (optimal weights, calibrated thresholds, and formal disagreement logging).

9.2 Second Opinion Quantification

The clinical practice of seeking a second opinion is the simplest non-trivial instantiation of Theorem 1: $M = 2$.

Example 9.1 (Second Opinion for Cancer Diagnosis)**.** A 62-year-old patient presents with a lung nodule on CT. Radiologist A ($\text{Sens}_A = 0.88, \text{Spec}_A = 0.92$) diagnoses malignancy. Radiologist B ($\text{Sens}_B = 0.85, \text{Spec}_B = 0.94$) independently reviews the same images and also diagnoses malignancy.

With two agreeing opinions ($\hat{d}_A = \hat{d}_B = 1$):

- The posterior probability of true malignancy increases from π_1 (pre-test probability) to close to 1.
- The probability that both agree on a false positive is: $\mathbb{P}(\text{both FP}) = (1 - \text{Spec}_A)(1 - \text{Spec}_B) = 0.08 \cdot 0.06 = 0.0048$.
- The effective $M = 2/(1 + \rho_{AB})$. If $\rho_{AB} = 0.3$ (both radiologists), $M = 1.54$. If they were from different specialties (radiologist + pulmonologist), $\rho \approx 0.15$, $M = 1.74$.

Disagreement case: If Radiologist A says malignant and Radiologist B says benign, the discrepancy signal $\delta_{\text{Yajie}} = 0.5$ (maximum). By Theorem 3, the source of disagreement — A over-calls, B under-calls, or the case is genuinely ambiguous — is unidentifiable without additional information (biopsy, follow-up imaging, multi-disciplinary review). The protocol mandates escalation to higher M and additional modalities K.

9.3 Screening False Positives vs. Rare Disease

Population screening programs — mammography, colonoscopy, newborn metabolic screening — operate in the rare-disease regime ($\pi_1 \ll 1$). Theorem 3 explains why false positives dominate in these settings.

Example 9.2 (Mammography Screening)**.** Breast cancer prevalence in screening population: $\pi_1 \approx 0.005$ (0.5%). Mammography: $\text{Sens} \approx 0.85, \text{Spec} \approx 0.90$.

For a positive mammogram:

$$\text{PPV} = \frac{\text{Sens} \cdot \pi_1}{\text{Sens} \cdot \pi_1 + (1 - \text{Spec})(1 - \pi_1)} = \frac{0.85 \cdot 0.005}{0.85 \cdot 0.005 + 0.10 \cdot 0.995} \approx 0.041. \tag{83}$$

Only 4.1% of positive screening mammograms represent true cancer — 95.9% are false positives. This is a direct consequence of Theorem 3: when π_1 is small, a single diagnostic modality cannot distinguish true positives from false positives.

SCX remedy: Add independent modalities ($K > 1$):

- Add diagnostic ultrasound ($\text{Sens} = 0.80, \text{Spec} = 0.85$): if both positive, $\text{PPV} \approx 0.19$ ($4.6\times$ improvement).
- Add MRI ($\text{Sens} = 0.92, \text{Spec} = 0.88$): if all three positive, $\text{PPV} \approx 0.52$.
- Add biopsy/pathology ($\text{Sens} \approx 0.98, \text{Spec} \approx 0.99$): $\text{PPV} \to 1$ for biopsy-positive cases.

This cascading diagnostic workflow — screening $\to$ diagnostic imaging $\to$ biopsy — is precisely a multi-modal Yajie consensus with sequentially increasing specificity, as formalized in Theorem 2.

9.4 AI-Assisted Diagnosis Under SCX Audit

AI diagnostic systems (deep learning for radiology, dermatology, pathology) raise a critical SCX question: **does an AI system count as an expert, or is it a tool used by experts?**

Proposition 9.3 (AI as an Adjacent Expert)**.** *An AI diagnostic system f_{AI} trained on dataset ${}_{AI}$ can serve as an expert in the* Yajie *consensus protocol if and only if:*

1. *Its training data ${}_{AI}$ is independent of the training data (clinical experience) of other physicians in the consensus panel (Assumption 3.3).*
2. *Its error correlation $\rho_{AI,m}$ with each human physician m is $\leq \rho_{\max}$ — i.e., it makes different errors than humans.*
3. *Its sensitivity and specificity are estimable from held-out data disjoint from both training and the current patient cohort (Assumption 3.8).*

When these conditions hold, the AI system adds one effective expert to the panel, increasing M. When the AI was trained on the same data distribution as the physicians' training (same hospital, same population), $\rho_{AI,m}$ may be high, reducing the added effective multiplicity.

9.5 Telemedicine and Distributed Diagnosis

Telemedicine platforms that connect patients to multiple remote physicians naturally create multi-expert diagnostic settings. The Yajie protocol can be implemented as a software layer:

1. Patient uploads medical data (imaging, lab results, history).
2. M physicians independently review and diagnose via the platform.
3. Platform computes Yajie consensus and discrepancy signals.
4. If δ_{Yajie} exceeds a threshold, the case is escalated to a specialist panel (higher M, additional modalities).
5. All diagnoses and outcomes are permanently stored in Spring memory for physician calibration and quality improvement.

10 Experimental Protocol and Empirical Validation

We specify the experimental protocol for validating Theorems 1–3 in clinical settings. We stress that this paper provides the mathematical framework; the empirical validation is specified for future work.

10.1 Data Requirements

(i) **Patient cohort:** $N \geq 1000$ patients with ground-truth diagnoses established by either (a) surgical pathology (gold standard), (b) long-term follow-up ($\geq$ 12 months), or (c) expert panel consensus with at least 3 independent specialists.

(ii) **Physician panel:** $M \geq 5$ physicians from distinct specialties, each providing independent diagnoses blinded to other physicians' opinions.

(iii) **Multi-modal data:** For each patient, collect all $K \geq 3$ diagnostic modality signals independently.

(iv) **Calibration set:** 200 historical cases per physician with ground-truth outcomes, used to estimate $\hat{\text{Sens}}_m, \hat{\text{Spec}}_m$.

10.2 Validation Metrics

(i) **Theorem 1 validation:** For each $m \in \{1, \dots, M\}$, compute the empirical misdiagnosis rate of the majority-vote consensus among subsets of m physicians. Compare to the bound $\exp(-2m\Delta^2)$. Plot $\log \varepsilon_{\text{mis}}$ vs. m — should be approximately linear with slope $-2\Delta^2$.

(ii) **Theorem 2 validation:** For each $k \in \{1, \dots, K\}$, compute the empirical misdiagnosis rate of the weighted consensus among subsets of k modalities. Compare to the bound $\exp(-k\bar{\Delta}^2/2)$. Verify that optimal weights $\omega_k = \ln(\text{DOR}_k)$ outperform uniform weights.

(iii) **Theorem 3 validation:** Stratify the patient cohort by disease prevalence. For each prevalence stratum, compute the empirical false negative identification rate as a function of M. Verify that for $\pi_1 \leq 10^{-3}$, $M \geq 5$ independent modalities are required to achieve $\mathbb{P}(\text{detect FN}) \geq 0.95$.

(iv) **Yajie consensus quality:** Compare the YAJIE weighted consensus against (a) simple majority vote, (b) single best physician, (c) random physician. Report AUC, sensitivity, specificity, and Net Reclassification Improvement (NRI).

10.3 Correlation Estimation Protocol

The effective multiplicity M depends critically on the inter-physician error correlation $\bar{\rho}$. We estimate $\rho_{mm'}$ via:

1. Collect N_{cal} calibration cases with known ground truth.
2. For each physician pair (m, m'), construct the 2×2 contingency table of errors:

$$\rho_{mm'} = \frac{n_{11} \cdot n_{00} - n_{10} \cdot n_{01}}{\sqrt{(n_{11} + n_{10})(n_{01} + n_{00})(n_{11} + n_{01})(n_{10} + n_{00})}}, \tag{84}$$

   where $n_{ab} = \#\{\text{cases where physician } m \text{ error } = a, \text{ physician } m' \text{ error } = b\}$.
3. Bootstrap confidence intervals: resample calibration cases with replacement $B = 1000$ times, recompute $\bar{\rho}$, report 95% CI.

11 Limitations and Assumption Verification

We state the limitations honestly. Every theorem in this paper holds under the eight assumptions of Section 3. Below we identify where these assumptions may fail in clinical practice.

[L1] **Conditional independence across modalities (3.2):** Imaging, lab, and pathology are physically distinct measurements, but they may be correlated through unobserved confounders (e.g., shared contrast agent effects, systemic inflammation affecting multiple lab values, tissue preparation artifacts affecting both histology and immunohistochemistry). This correlation reduces K. *Mitigation:* Estimate $\bar{\rho}_K$ from calibration data and use the effective count K rather than K.

[L2] **Inter-physician correlation estimation (3.3):** Estimating $\rho_{mm'}$ requires ground-truth diagnoses, which are themselves noisy. Using expert panel consensus as ground truth introduces circularity — the same physicians whose correlation we measure may constitute the ground-truth panel. *Mitigation:* Use surgical pathology or long-term follow-up as independent ground truth.

[L3] **Diagnostic margin Δ is case-dependent (3.4):** The margin Δ varies across cases — clear cases have $\Delta \approx 0.5$, ambiguous cases have $\Delta \approx 0$. Using a single Δ for all cases overestimates detection probability for easy cases and underestimates it for difficult cases. *Mitigation:* Stratify by case difficulty, estimated via inter-physician agreement rate (high agreement $\implies$ high Δ).

[L4] **Prevalence stability (3.5):** Disease prevalence shifts across populations (geographic, demographic, temporal). A diagnostic protocol calibrated for π_1 in one population may be miscalibrated in another. *Mitigation:* SPRING gating detects prevalence shifts by monitoring the ratio of positive to negative diagnoses over time.

[L5] **Physician sensitivity/specificity drift (3.8):** Physicians' diagnostic accuracy changes over time — learning improves it, fatigue degrades it, new guidelines shift decision boundaries. Static $\widehat{\mathrm{Sens}}_m, \widehat{\mathrm{Spec}}_m$ become stale. *Mitigation:* Continuous SPRING monitoring of physician outcomes updates weights ω_m adaptively.

[L6] **Theorem 3 resolution requirement:** For ultra-rare diseases ($\pi_1 < 10^{-6}$), the required M for Theorem 3 resolution may exceed the number of available specialists worldwide. The mathematical guarantee exists, but the practical implementation is impossible. *Honesty:* The theorem identifies a fundamental limitation, not a practical solution. For such diseases, diagnosis must rely on pathognomonic findings (signs/symptoms that are uniquely associated with the disease) rather than statistical consensus.

12 Discussion

12.1 Why Medicine Has Multi-Expert Intuition But No Formal Theory

Medicine has practiced multi-expert diagnosis for centuries — consulting colleagues, grand rounds, tumor boards, morbidity and mortality conferences. Yet no formal mathematical theory connects the number of physicians M to the misdiagnosis probability ε. This gap has practical consequences:

1. **No guidance on how many opinions to seek.** Is $M = 2$ sufficient? $M = 5$? The medical literature provides empirical evidence (second opinions change diagnosis in 10–62% of cases) but no formula.
2. **No quantification of diminishing returns.** When does adding another physician stop reducing misdiagnosis probability? When $\bar{\rho}$ is high, M saturates quickly.
3. **No formal justification for tumor board size.** Why 5–10 specialists and not 3 or 20? Theorem 1 provides the answer: compute M and the target ε.

12.2 The Spring Permanent Memory in Medicine

The Spring permanent memory mechanism has a natural clinical analog: the electronic health record (EHR). However, current EHRs violate Spring's monotonicity guarantee (Theorem Spring-1 from the core SCX framework): data can be deleted, modified, or lost. A Spring-compliant medical record would:

1. **Append-only:** Every diagnosis, test result, and clinical note is permanently recorded with timestamp and physician identity.
2. **Immutable audit trail:** Changes to a diagnosis (e.g., initial diagnosis revised after further testing) are recorded as new entries linked to the original, not as overwrites.
3. **Cumulative quality tracking:** Each physician's cumulative sensitivity and specificity are continuously updated, enabling real-time Yajie weight calibration.

12.3 The Cost of Multi-Expert Diagnosis

The primary objection to multi-expert diagnosis is cost: M physicians cost M times as much as one. The SCX framework reframes this as an optimization problem:

$$M^* = \arg\min_{M \in \mathbb{N}} \Big\{ M \cdot c_{\text{physician}} + \varepsilon_{\text{mis}}(M) \cdot c_{\text{misdiagnosis}} \Big\}, \tag{85}$$

where $c_{\text{physician}}$ is the cost per physician consultation and $c_{\text{misdiagnosis}}$ is the expected cost of a misdiagnosis (including litigation, delayed treatment, unnecessary procedures, and human life).

For a disease where $c_{\text{misdiagnosis}} \gg c_{\text{physician}}$ (e.g., misdiagnosed cancer $\to$ death), the optimal M^* is large. For a disease where $c_{\text{misdiagnosis}} \approx c_{\text{physician}}$ (e.g., mild self-limiting condition), $M^* = 1$ is optimal.

12.4 Regulatory Implications

The SCX framework implies that medical licensing, hospital privileging, and diagnostic device approval should incorporate M-parameter declaration:

1. **Diagnostic AI systems** should declare M — the number of independent human experts whose consensus the AI's output is equivalent to.
2. **Clinical practice guidelines** should specify M_{min} for high-stakes diagnostic decisions.
3. **Malpractice standards** should incorporate M: a physician who fails to seek a second opinion when Theorem 1 indicates $M_{\text{min}} > 1$ may be considered negligent.

13 Related Work

The literature on diagnostic accuracy is vast. We highlight the mathematical connections most relevant to the SCX framework.

Diagnostic accuracy meta-analysis. Systematic reviews of diagnostic accuracy (e.g., Cochrane DTA reviews) estimate pooled sensitivity and specificity across studies. These provide $\widehat{\text{Sens}}, \widehat{\text{Spec}}$ estimates that serve as inputs to Theorems 1–3, but they do not consider multi-expert consensus error bounds.

Second opinion studies. Empirical studies of second opinions [6, 5] document opinion-change rates but do not derive the mathematical relationship between M, $\bar{\rho}$, and misdiagnosis probability.

Tumor board outcomes. Studies of multidisciplinary team (MDT) meetings [7, 8] report diagnosis or management changes in 25–52% of cases but do not formalize the consensus mechanism or provide error bounds.

Ensemble methods in medical AI. AI research has produced ensemble diagnostic models (e.g., multiple CNNs for radiology, ensemble of gradient-boosted trees for clinical prediction).

These implicitly use $M > 1$ experts but: (a) the experts share the same architecture and training data ($\bar{\rho}$ is high), (b) the M parameter is never declared, and (c) error detection bounds are not provided.

Probabilistic diagnosis. Bayesian diagnostic reasoning [9] computes posterior disease probabilities given test results. This is the single-modality, single-physician version of our framework. The SCX extension is the multi-expert, multi-modality generalization with explicit error bounds.

Condorcet jury theorem. The Condorcet jury theorem (1785) proves that majority vote among independent voters converges to the correct decision as $M \to \infty$. Theorem 1 generalizes this with finite-M Hoeffding bounds, correlation correction via M, and the diagnostic margin Δ.

14 Conclusion

We have formalized medical diagnosis as a multi-expert audit problem under the SCX framework. The three theorems provide:

1. **Theorem 1 (Multi-Physician Misdiagnosis Detection Theorem):** The probability that M physicians collectively misdiagnose decays as $\exp(-2M\Delta^2)$. Required physician count for target misdiagnosis rate ε is $M \geq \ln(1/\varepsilon)/(2\Delta^2)\cdot(1+(M-1)\bar{\rho})$. For independent physicians ($\bar{\rho} = 0, \Delta = 0.3, \varepsilon = 0.05$), $M \geq 17$. For correlated physicians ($\bar{\rho} = 0.5$), M must be proportionally larger. The second opinion ($M = 2$) provides detectable but limited error reduction.
2. **Theorem 2 (Multi-Modal Expert Consensus Theorem):** When K diagnostic modalities provide conditionally independent signals, the Yajie weighted consensus with optimal weights $\omega_k = \ln(\mathrm{DOR}_k)$ achieves misdiagnosis probability $\leq \exp(-K\bar{\Delta}^2/2)$. The standard clinical workflow — imaging $\to$ lab $\to$ pathology — is a sequential implementation of this consensus. The cost-optimal threshold $\tau^* = \ln(c_{\mathrm{FP}}(1-\pi_1)/(c_{\mathrm{FN}}\pi_1))$ balances false negative and false positive costs.
3. **Theorem 3 (False Negative and Rare Disease Unidentifiability Theorem):** When disease prevalence $\pi_1 \to 0$, a single diagnostic modality cannot distinguish between true disease absence and false negative. The mutual information $I(\hat{d}; d) \to 0$. Resolution requires $M > \ln(1/\varepsilon)/(2\Delta^2)$ independent modalities. This explains the dominance of false positives in population screening (PPV $\approx 4\%$ for mammography) and quantifies the benefit of cascade testing protocols.

The Yajie protocol operationalizes these theorems into a structured diagnostic workflow suitable for tumor boards, multidisciplinary teams, telemedicine platforms, and AI-assisted diagnosis. The Cercis score $S = Q + \eta N$ provides a single auditable metric for diagnostic quality, and the Spring permanent memory enables continuous physician calibration and retrospective error analysis.

Medicine is the domain where audit matters most. A misdiagnosis is not a misclassification on a benchmark — it is a human life misjudged. The SCX framework does not replace clinical judgment. It provides the mathematical structure to certify it.

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